Engineering Economics

Chapter 1: Introduction — Questions & Answers

Question 1

> ***Define Engineering Economics. Why is the study of economics essential for engineers? Explain the role of engineers in decision making and state the principles of Engineering Economics.***

Answer:

Definition of Engineering Economics:

Engineering Economics is the application of economic principles and analytical methods to engineering projects and business decisions. It provides a systematic framework for evaluating the financial feasibility of engineering alternatives and selecting the most cost-effective solution.

Importance of Economics for Engineers:

Engineers frequently make decisions involving large capital expenditures. Without economic knowledge:

- They cannot assess whether a project is financially viable.
- They cannot compare alternatives differing in cost, life span, and performance.
- They risk recommending technically correct but economically poor solutions.
- They cannot estimate payback periods or life-cycle costs.

> ***Example:*** *An engineer choosing between a high-cost pipe (30-year life) vs. a cheap pipe (10-year life) must use economic analysis to determine which is truly more cost-effective over the project life.*

Role of Engineers in Decision Making:

1. Problem Identification: Clearly define the technical and economic problem.
2. Generating Alternatives: Develop all feasible engineering solutions.
3. Evaluating Alternatives: Apply economic methods like PW, AW, IRR, and B/C ratio to each option.
4. Selecting the Best Alternative: Recommend the option best meeting technical, economic, and social criteria.
5. Implementation: Execute the chosen solution.
6. Monitoring and Review: Track performance and revise decisions as new information emerges.

Principles of Engineering Economics:

1. Use a Consistent Viewpoint: Evaluate all cash flows from a defined perspective (owner, firm, or society).
2. Use a Common Unit of Measurement: Convert all costs and benefits into monetary units.
3. Consider All Relevant Criteria: Include economic AND non-economic factors (environmental, social, ethical).
4. Make Uncertainty Explicit: Identify and quantify risks in estimates.
5. Revisit Decisions: Be willing to revise decisions as better information becomes available.
6. Separate Sunk Costs: Do not allow irrecoverable past costs to influence future decisions.
7. Focus on Differences: Only differences between alternatives matter when comparing options.
8. Compare over a Common Study Period: Evaluate all alternatives over the same time horizon.
9. People Are More Important Than Numbers: Human values and ethics must guide every engineering decision.

Question 2

> ***What do you mean by sunk cost and opportunity cost? Explain with examples.***

Answer:

Sunk Cost:

A sunk cost is a past cost that has already been incurred and cannot be recovered regardless of any future decision or action. Since it cannot be changed, it should NOT be included in future economic decision-making. Only future incremental costs and benefits should matter.

> ***Example:*** *A company spent Rs. 5,00,000 on a feasibility study for a project later found unviable. This Rs. 5,00,000 is a sunk cost. Whether the company continues or abandons the project, this money is gone. It must be excluded when deciding whether to proceed.*

Key Rule: Sunk costs are irrelevant to future engineering decisions. Always focus only on future costs and benefits.

Opportunity Cost:

Opportunity cost is the value of the best alternative forgone when a particular decision is made. It represents the benefit sacrificed by choosing one option over another. Every decision has an opportunity cost even if no cash payment is made.

> ***Example:*** *If an engineer invests Rs. 1,00,000 in machinery instead of depositing it in a bank at 10% per year, the opportunity cost is Rs. 10,000/year — the foregone interest income.*
>
> ***Example:*** *If a factory uses its floor space for Product A instead of Product B, the opportunity cost is the profit that could have been earned from Product B.*

Key Rule: Opportunity costs must be recognized and included when comparing engineering alternatives, even if they do not appear explicitly in financial records.

Question 3

> ***Define and draw a Cash Flow Diagram. Explain its components with a suitable numerical example.***

Answer:

Cash Flow Diagram (CFD):

A Cash Flow Diagram is a graphical tool representing the timing and magnitude of all cash receipts (inflows) and disbursements (outflows) associated with a project over its life. It helps engineers visualize all monetary transactions, making economic analysis clearer and more systematic.

Components of a Cash Flow Diagram:

1. Time Axis (Horizontal Line): Represents time divided into equal intervals (usually years). Period 0 = present.
2. Upward Arrows (↑): Cash INFLOWS — revenues, salvage value, savings (positive cash flows).
3. Downward Arrows (↓): Cash OUTFLOWS — initial investment, operating costs, loan payments (negative cash flows).
4. Arrow Height: Proportional to the magnitude of the cash flow amount.
5. Perspective: Drawn from the viewpoint of the decision-maker (investor or project owner).

Numerical Example:

A machine is purchased for Rs. 1,00,000 now (year 0). It generates annual income of Rs. 30,000/year for 5 years and has a salvage value of Rs. 20,000 at end of year 5.

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                    30K   30K   30K   30K   50K
                     ↑     ↑     ↑     ↑     ↑
                    |-------|-----|-----|-----|-----|
0      1     2     3     4     5   (years)
                                ↓
                            1,00,000
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Note: Arrow at year 5 = Rs. 30,000 (income) + Rs. 20,000 (salvage) = Rs. 50,000. Downward arrow at year 0 shows the initial investment cost.

Question 4

> ***Differentiate between Consumer Goods and Producer Goods.***

Answer:

Consumer Goods:

Consumer goods are final products manufactured for direct consumption by individuals or households. They satisfy human wants and needs directly without further processing. They may be durable (lasting multiple years) or non-durable (consumed quickly).

- Durable Consumer Goods: Refrigerator, television, automobile.
- Non-durable Consumer Goods: Food, clothing, soap, fuel.

Producer Goods (Capital Goods):

Producer goods are items used in the production of other goods or services. They are not consumed directly by end users but serve as inputs in manufacturing or industrial processes. They are also called capital goods.

> ***Example:*** *CNC machines, factory buildings, industrial pumps, conveyors, power plants, raw materials for production.*

Key Differences:

- Purpose: Consumer goods satisfy direct human needs; Producer goods are used to manufacture other products or services.
- End User: Consumer goods are used by households; Producer goods are used by firms and industries.
- Economic Analysis: Consumer goods involve demand/utility analysis; Producer goods require investment evaluation, depreciation, and ROI calculations.
- Engineering Economics Relevance: Engineering economics primarily deals with producer/capital goods — investment decisions, replacement analysis, and life-cycle cost evaluation of industrial equipment.

Question 5

> ***Explain the scope and limitations of Engineering Economics.***

Answer:

Scope of Engineering Economics:

1. Cost Analysis and Estimation: Fixed, variable, direct, indirect, overhead, and life-cycle cost evaluation.
2. Time Value of Money: Present worth, future worth, and equivalence of cash flows at different time periods.
3. Investment Analysis: Evaluating capital projects using PW, AW, FW, IRR, and Benefit-Cost Ratio.
4. Depreciation and Tax Analysis: Computing depreciation (SL, DB, MACRS) and its effect on taxable income and after-tax cash flows.
5. Replacement Analysis: Determining the optimal time to retire and replace aging equipment.
6. Break-even Analysis: Finding the production volume at which total revenue equals total cost.
7. Risk and Sensitivity Analysis: Evaluating the impact of uncertainties in estimates on project outcomes.
8. Public Project Evaluation: Benefit-Cost Analysis for government and infrastructure projects.
9. Inflation Analysis: Adjusting economic analyses to account for changing price levels over time.

Limitations of Engineering Economics:

1. Quantitative Bias: Primarily handles measurable monetary factors. Social, ethical, and political impacts are difficult to quantify.
2. Estimation Inaccuracies: Future costs, revenues, and service lives are estimated values. Errors can significantly distort recommendations.
3. Assumes Rational Behavior: Models assume decision-makers always act rationally, which may not hold in practice.
4. Data Dependency: Reliable analysis requires accurate historical and market data that may not always be available.
5. Changing Conditions: Inflation, tax laws, and interest rates change unpredictably, reducing accuracy of long-term projections.
6. Ignores Intangible Benefits: Factors like employee morale, brand reputation, and community goodwill are not easily captured in standard models.

Question 6

> ***Explain the Law of Demand and Demand Curve. What are the factors affecting demand?***

Answer:

Law of Demand:

The Law of Demand states that, ceteris paribus (all other factors remaining constant), as the price of a good increases, the quantity demanded decreases; and as the price decreases, the quantity demanded increases. This establishes an inverse (negative) relationship between price and quantity demanded.

$$QD = a - bP \quad (a = \text{intercept}, b = \text{slope coefficient}, P = \text{price})$$

Demand Curve:

The demand curve graphically shows the relationship between price (Y-axis) and quantity demanded (X-axis). It slopes downward from left to right, consistent with the Law of Demand. A movement ALONG the demand curve results from a price change; a SHIFT of the entire demand curve results from non-price factors.

Factors Affecting Demand (Non-Price Determinants):

1. Income of Consumers: Higher income raises demand for normal goods and lowers demand for inferior goods.

2. Price of Related Goods: Rise in substitute prices increases demand; rise in complementary goods prices decreases demand.
3. Consumer Tastes and Preferences: A positive shift in tastes increases demand for that good.
4. Number of Buyers: More buyers in the market increases total market demand.
5. Consumer Expectations: If consumers expect future price increases, current demand rises as people buy more now.
6. Advertising and Marketing: Effective promotion creates awareness and desire, shifting demand rightward.
7. Seasonal Factors: Demand for certain goods (raincoats, heaters) fluctuates with seasons.

Question 7

> ***Differentiate between Microeconomics and Macroeconomics.***

Answer:

Microeconomics:

Microeconomics studies the economic behavior of individual decision-making units — consumers, households, firms, and industries. It analyzes how these units interact in specific markets to determine prices and quantities of individual goods and services.

Key topics: Demand and supply, price determination, consumer behavior, production and cost theory, market structures, and factor pricing.

> ***Example:*** *How does a rise in raw material costs affect the output decision of a manufacturing firm?*

Macroeconomics:

Macroeconomics studies the economy as a whole. It analyzes aggregate economic variables and the overall performance of national and global economies.

Key topics: GDP, national income, aggregate demand/supply, inflation, unemployment, fiscal policy, monetary policy, balance of payments, and economic growth.

> ***Example:*** *How does a government budget deficit affect national employment and the inflation rate?*

Key Differences:

- Scope: Micro = individual/firm level; Macro = national/global economy level.
- Focus: Micro = price and output of specific goods; Macro = total output (GDP), overall price level, employment.
- Tools: Micro = demand/supply curves, indifference curves; Macro = IS-LM models, aggregate demand/supply.
- Problems Addressed: Micro = resource allocation, pricing, market structure; Macro = inflation, unemployment, growth.
- Engineering Relevance: Microeconomics is directly applied in project evaluation; Macroeconomics sets the investment and policy environment.

Question 8

> ***Explain the concept of Elasticity of Demand and describe its different types.***

Answer:

Elasticity of Demand:

Elasticity of demand measures the degree of responsiveness of the quantity demanded of a good to a change in one of its determinants. It is expressed as the ratio of the percentage change in quantity demanded to the percentage change in the determining variable.

1. Price Elasticity of Demand (PED):

Measures the percentage change in quantity demanded resulting from a percentage change in the price of the good.

PED = (% Change in Qty Demanded) / (% Change in Price)

- Perfectly Elastic (PED = ∞): Any price increase causes demand to fall to zero. Horizontal demand curve.
- Elastic (PED > 1): Demand changes more than proportionally to price. Common for luxury goods.
- Unit Elastic (PED = 1): Demand changes proportionally equal to the price change.
- Inelastic (PED < 1): Demand changes less than proportionally. Common for necessities like food and fuel.
- Perfectly Inelastic (PED = 0): Demand is completely unresponsive to price. Vertical demand curve. E.g., life-saving insulin.

2. Income Elasticity of Demand (YED):

Measures the responsiveness of demand to changes in consumer income.

YED = (% Change in Qty Demanded) / (% Change in Income)

- Positive YED: Normal goods — demand rises as income increases.
- Negative YED: Inferior goods — demand falls as income rises.
- YED > 1: Luxury goods — demand rises faster than income growth.

3. Cross Elasticity of Demand (XED):

Measures how demand for one good responds to a change in the price of another related good.

XED = (% Change in Qty of Good A) / (% Change in Price of Good B)

- Positive XED: Substitute goods (e.g., Pepsi and Coca-Cola) — price rise of one increases demand for the other.
- Negative XED: Complementary goods (e.g., cars and petrol) — price rise of one decreases demand for the other.
- Zero XED: Unrelated goods — no cross-price relationship.